

National University of Computer and Emerging Sciences

FAST School of Computing

Spring-2022

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Question 1 [5+5+10=20 Marks]

- I. Process the SR Latch given below for all possible cases and fill the table accordingly?

Note: Apply given SR values as an input to the circuit & calculate Q_{T+1} & $\overline{Q}_{T+1}$ for all case given below.

NO marks for DIRECT ANSWERS

CIRCUIT:

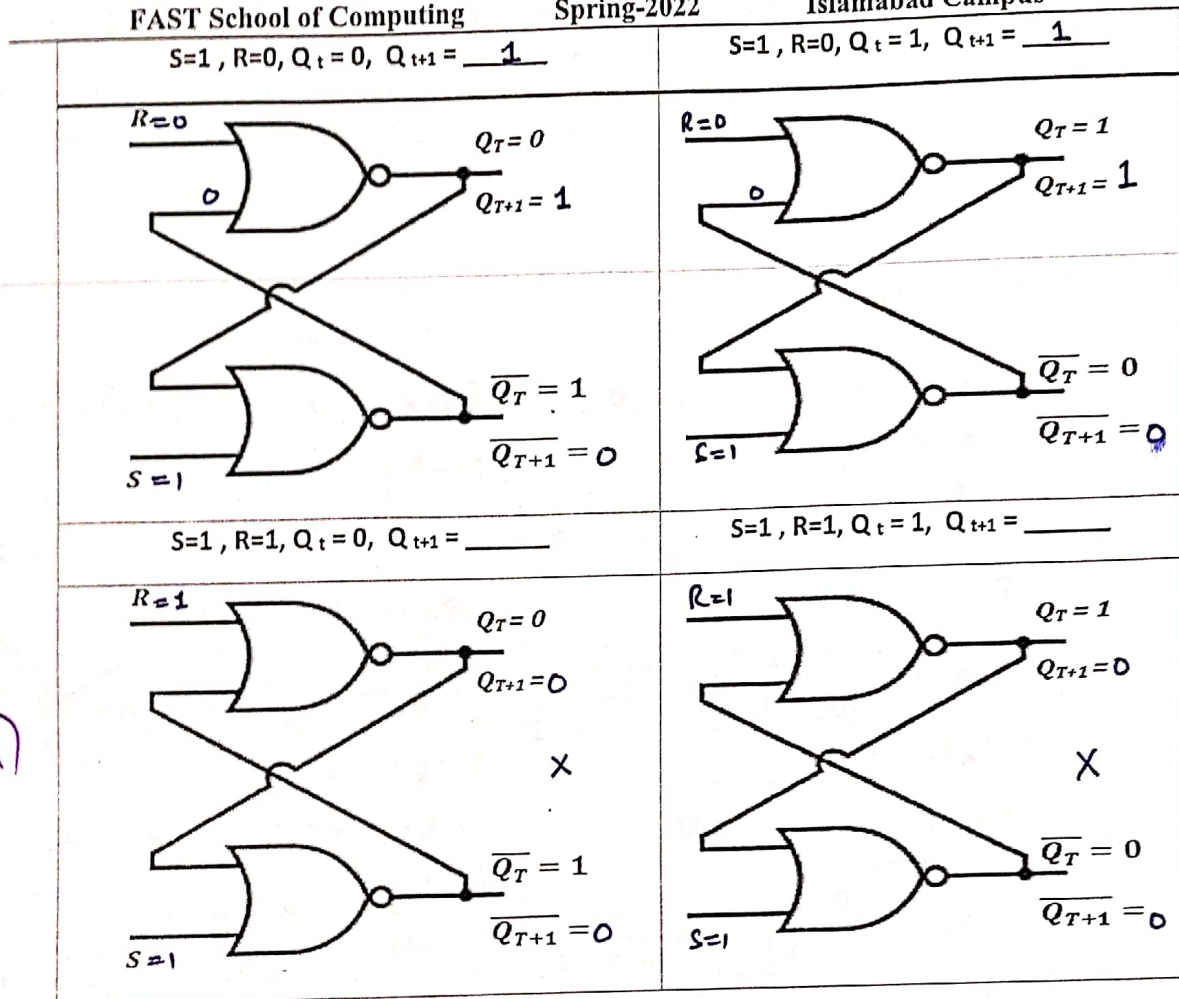
Case-I ($Q_t = 0$)	Case-II ($Q_t = 1$)
$S=0, R=0, Q_t = 0, Q_{t+1} = \underline{0}$ $Q_T = 0$ $Q_{T+1} = 0$ $\overline{Q}_T = 1$ $\overline{Q}_{T+1} = 1$	$S=0, R=0, Q_t = 1, Q_{t+1} = \underline{1}$ $Q_T = 1$ $Q_{T+1} = 1$ $\overline{Q}_T = 0$ $\overline{Q}_{T+1} = 0$
$S=0, R=1, Q_t = 0, Q_{t+1} = \underline{0}$ $Q_T = 0$ $Q_{T+1} = 0$ $\overline{Q}_T = 1$ $\overline{Q}_{T+1} = 1$	$S=0, R=1, Q_t = 1, Q_{t+1} = \underline{0}$ $Q_T = 1$ $Q_{T+1} = 0$ $\overline{Q}_T = 0$ $\overline{Q}_{T+1} = 1$

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(1) - 0.25

S	R	Q_{T+1}	$\overline{Q_{T+1}}$
0	0	$Q(t)$	$[Q(t)]'$
0	1	0	1
1	0	1	0
1	1	0(x)	0(x)

undefined / Not allowed

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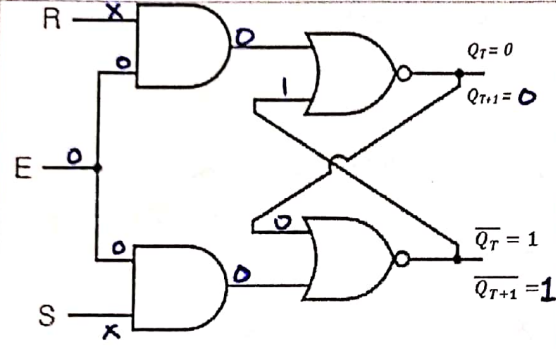
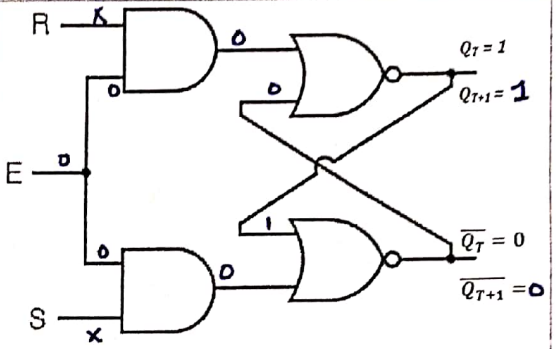
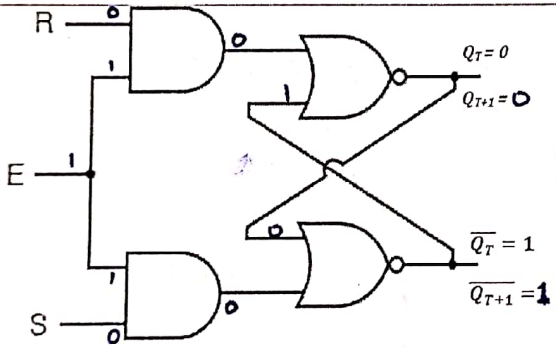
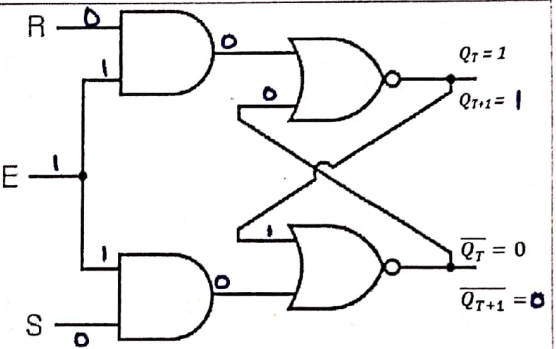
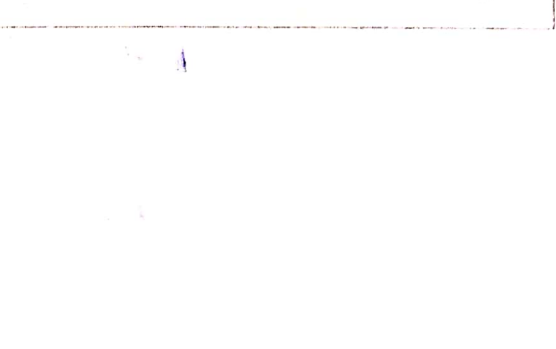
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- ii. Process the **Gated SR Latch** given below for all possible cases and fill the table accordingly?

Note: Apply given **SR** values and **E(Enable)** as an input to the **Gated SR** circuit & calculate Q_{T+1} & $\overline{Q}_{T+1}$ for all case given below. You are not supposed to process **SR-Latch** use the table achieved in part-I

NO marks for DIRECT ANSWERS

CIRCUIT:

Case-I ($Q_t = 0$)	Case-II ($Q_t = 1$)
$E=0, S=\underline{x}, R=\underline{x}, Q_t=0, Q_{t+1}=\underline{0}$  $Q_T=0$ $Q_{T+1}=0$ $\overline{Q}_T=1$ $\overline{Q}_{T+1}=1$	$E=0, S=\underline{x}, R=\underline{x}, Q_t=1, Q_{t+1}=\underline{1}$  $Q_T=1$ $Q_{T+1}=1$ $\overline{Q}_T=0$ $\overline{Q}_{T+1}=0$
$E=1, S=0, R=0, Q_t=0, Q_{t+1}=\underline{0}$  $Q_T=0$ $Q_{T+1}=0$ $\overline{Q}_T=1$ $\overline{Q}_{T+1}=1$	$E=1, S=0, R=0, Q_t=1, Q_{t+1}=\underline{1}$  $Q_T=1$ $Q_{T+1}=1$ $\overline{Q}_T=0$ $\overline{Q}_{T+1}=0$
$E=1, S=0, R=1, Q_t=0, Q_{t+1}=\underline{0}$  $Q_T=0$ $Q_{T+1}=0$ $\overline{Q}_T=1$ $\overline{Q}_{T+1}=1$	$E=1, S=0, R=1, Q_t=1, Q_{t+1}=\underline{0}$  $Q_T=1$ $Q_{T+1}=0$ $\overline{Q}_T=0$ $\overline{Q}_{T+1}=1$

0-25 each
↑
(1)

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$E=1, S=1, R=0, Q_t=0, Q_{t+1} = \underline{1}$

$E=1, S=1, R=0, Q_t=1, Q_{t+1} = \underline{1}$

$E=1, S=1, R=1, Q_t=0, Q_{t+1} = \underline{\quad}$

$E=1, S=1, R=1, Q_t=1, Q_{t+1} = \underline{\quad}$

$E=1, S=1, R=1, Q_t=0, Q_{t+1} = \underline{\quad}$

$E=1, S=1, R=1, Q_t=1, Q_{t+1} = \underline{\quad}$

E	S	R	Q_{T+1}	$\overline{Q}_{T+1}$
0	x	x	$Q(t)$	$[Q(t)]'$
1	0	0	$Q(t)$	$[Q(t)]'$
1	0	1	0	1
1	1	0	1	0
1	1	1	0 (x)	0 (x)

undefined
Not Allowed
Invalid

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iii. Process the JK Latch given below for all possible cases and fill the table accordingly?

Note: Apply given JK values and E(Enable) as an input to the circuit & calculate Q_{t+1} & $\overline{Q}_{t+1}$ for all case given below You are not supposed to process SR-Latch use the table achieved in part-I.

CIRCUIT:

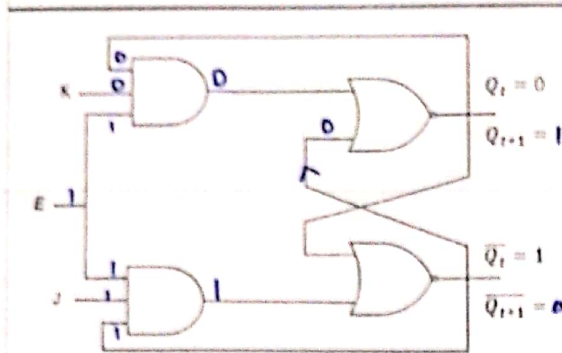
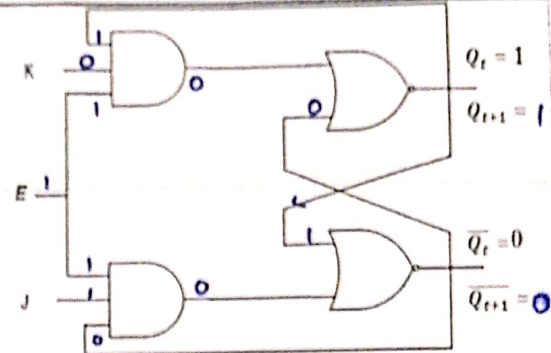
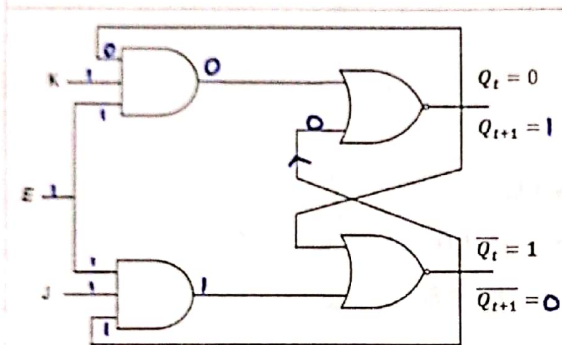
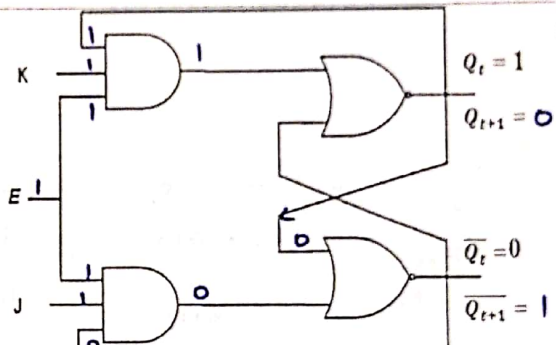
Case-I ($Q_t = 0$)	Case-II ($Q_t = 1$)
$E=0, J=\underline{x}, K=\underline{x}, Q_t=0, Q_{t+1}=\underline{0}$	$E=0, J=\underline{x}, K=\underline{x}, Q_t=1, Q_{t+1}=\underline{1}$
$E=1, J=0, K=0, Q_t=0, Q_{t+1}=\underline{0}$	$E=1, J=0, K=0, Q_t=1, Q_{t+1}=\underline{1}$
$E=1, J=0, K=1, Q_t=0, Q_{t+1}=\underline{0}$	$E=1, J=0, K=1, Q_t=1, Q_{t+1}=\underline{0}$

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 $E=1, J=1, K=0, Q_t=0, Q_{t+1} = \underline{1}$  $E=1, J=1, K=0, Q_t=1, Q_{t+1} = \underline{1}$  $E=1, J=1, K=1, Q_t=0, Q_{t+1} = \underline{1}$  $E=1, J=1, K=1, Q_t=1, Q_{t+1} = \underline{0}$ 

(2)

Enable	J	K	Q_{t+1}	$\overline{Q}_{t+1}$
0	X	X	$Q(t)$	$[Q(t)]'$
0.5 1	0	0	$Q(t)$	$[Q(t)]'$
.25 1	0	1	0	1
.25 1	1	0	1	0
.25 1	1	1	$[Q(t)]'$	$Q(t)$

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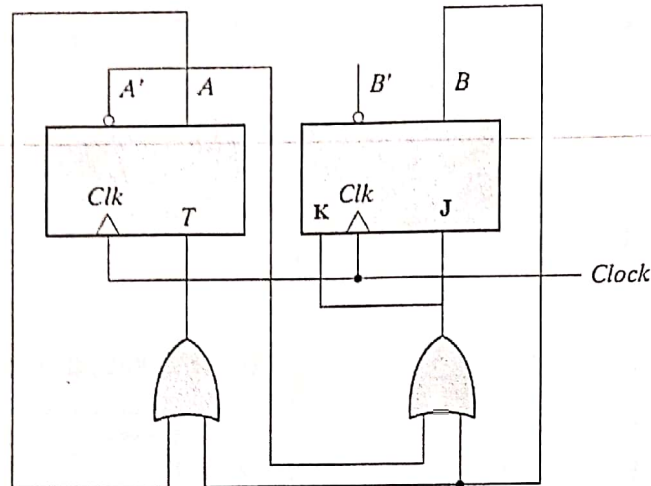
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Question 2 [5+10+5=20 Marks]

Derive the State Equation, state table and the state diagram of the sequential circuit given below.



State Equation

$$T_{A(t+1)} = A(t) + B(t) \Rightarrow T_A = A + B \rightarrow (1) \quad \checkmark$$

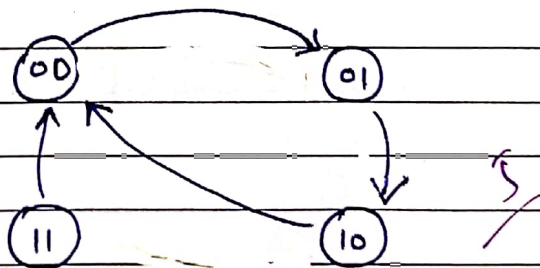
$$J_{B(t+1)} = B(t) + (A(t))' \Rightarrow J_B = B + A' \rightarrow (2) \quad \checkmark$$

$$K_{B(t+1)} = J_{B(t+1)} \Rightarrow K_B = B + A' \rightarrow (3) \quad \checkmark$$

State Table & State Diagram

T_A	A	B	$A(t+1)$	$B(t+1)$	J_B	K_B
0	0	0	0	1	1	1
1	0	1	1	0	1	1
1	1	0	0	0	0	0
1	1	1	0	0	1	1

$\rightarrow$ (2) entries
 put values in eq. (1) for $A(t+1)$ and values in (2) and (3) to find J_B and K_B . then use J_B and K_B to find $B(t+1)$



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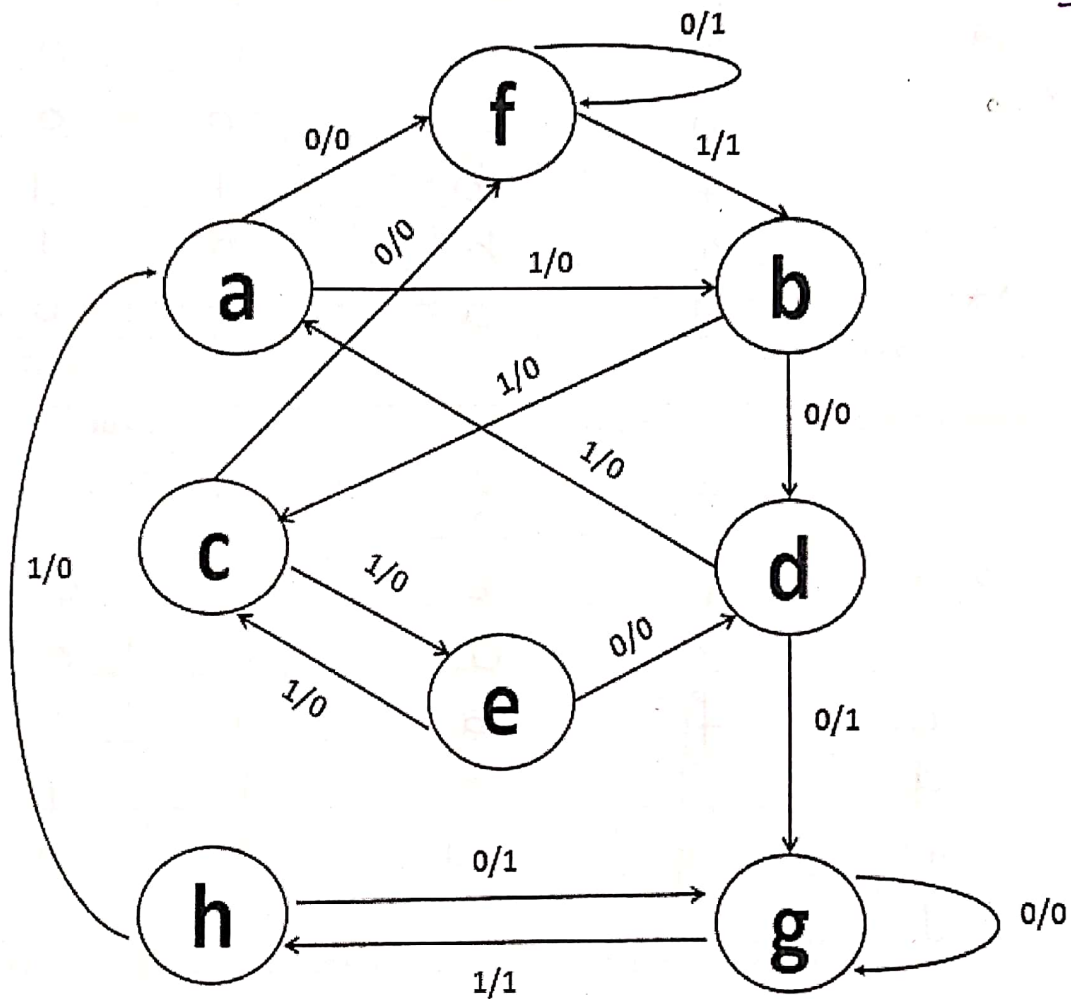
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Question 3 [3+7+7+3=20 Marks]

- i. For STATE DIAGRAM given below determine output where input sequence is 01110010011011 where starting state is a

INPUT	0	1	1	1	0	0	1	0	0	1	1	0	1
STATE	a	f	b	c	e	d	g	h	g	g	h	a	f
OUTPUT	0	1	0	0	0	1	1	1	0	1	0	0	1

State Diagram



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- ii. Complete State Table below for State Diagram give above?

Present State	Next State		Output	
	x = 0	x = 1	x = 0	x = 1
X a	f	b	0	0
• b	d	c	0	0
X c	f	e b	0	0 drop c
✱ d	g	a	1	0
• e	d	c	0	0 drop e
f	f	b	1	1
g	g	kd	0	1
✱ h	g	a	1	0 drop h

- iii. Generate Reduced State Table

Present State	Next State		Output	
	x = 0	x = 1	x = 0	x = 1
a	f	b	0	0
b	d	a	0	0
d	g	a	1	0
f	f	b	1	1
g	g	d	0	1

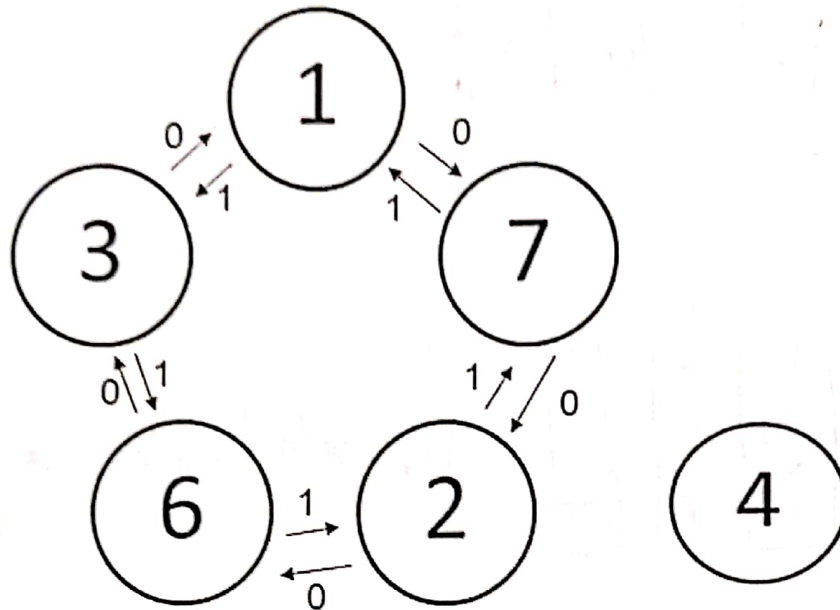
- iv. For Reduced State Table, determine output where input sequence is 01110010011, starting from state is a

INPUT	0	1	1	1	0	0	1	0	0	1	1	0	1
STATE	a	f	b	a	b	d	g	d	g	g	d	a	f
OUTPUT	0	1	0	0	0	1	1	1	0	1	0	0	1

→ States are diff b/c few states dropped
 → But output remains same

Question 4 [15 + 5=20 Marks]

Design the sequential circuit using D-Flip-flop specified by the state diagram given below? find self-correcting state for unused state 4?



State Table

Input	Present State			Next State			Output/Flip flops		
X	Q ₂	Q ₁	Q ₀	Q ₂	Q ₁	Q ₀	D ₂	D ₁	D ₀
0	0	0	0	X	X	X	X	X	X
0	0	0	1	1	1	1	1	1	1
0	0	1	0	1	1	0	1	1	0
0	0	1	1	0	0	1	0	0	1
0	1	0	0	X	X	X	X	X	X
0	1	0	1	X	X	X	X	X	X
0	1	1	0	0	1	1	0	1	1
0	1	1	1	0	1	0	0	1	0
1	0	0	0	X	X	X	X	X	X
1	0	0	1	0	1	1	0	1	1
1	0	1	0	1	1	1	1	1	1
1	0	1	1	1	1	0	1	1	0
1	1	0	0	X	X	X	X	X	X
1	1	0	1	X	X	X	X	X	X
1	1	1	0	0	1	0	0	1	0
1	1	1	1	0	0	1	0	0	1

D_2

$x \backslash Q_2 Q_1 Q_0$	00	01	11	10
00	X	1	0	1
01	X	X	0	0
11	X	X	0	0
10	X	0	1	1

$D_2 = x'Q_1' + xQ_2'Q_1 + Q_2'Q_1Q_0'$

D_1

$x \backslash Q_2 Q_1 Q_0$	00	01	11	10
00	X	1	0	1
01	X	X	1	1
11	X	X	0	X
10	X	1	1	1

(3)

$$D_1 = x'Q_1' + x'Q_2 + xQ_2' + Q_1Q_0'$$

D_0

$x \backslash Q_2 Q_1 Q_0$	00	01	11	10
00	X	1	1	0
01	X	X	0	1
11	X	1	0	1
10	X	X	1	0

(4)

$$D_0 = Q_1' + Q_2Q_0' + Q_2'Q_0$$

For 4(100) $x=0$. $\rightarrow$ put values in eqs of D_2, D_1 & D_0

Correcting State = 111 (7)

(2.5')

For 4(100) $x=1$

Correcting State = 001 (1)

(2.5')

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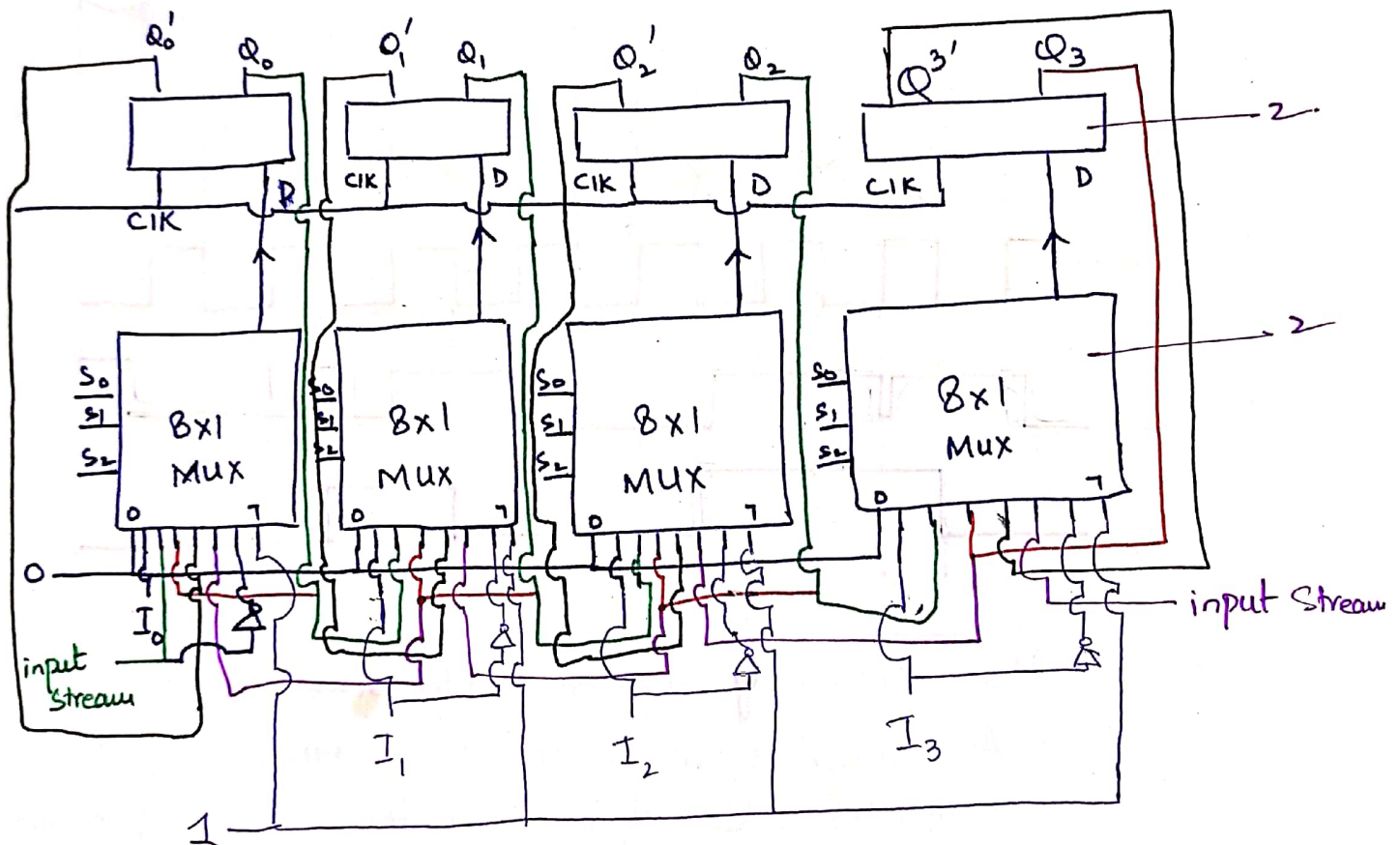
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Question 5 [20 Marks]

Draw the 4-bit Universal shift Register with four D flip-flops and four 8×1 multiplexers. The register operates according to the following function table.

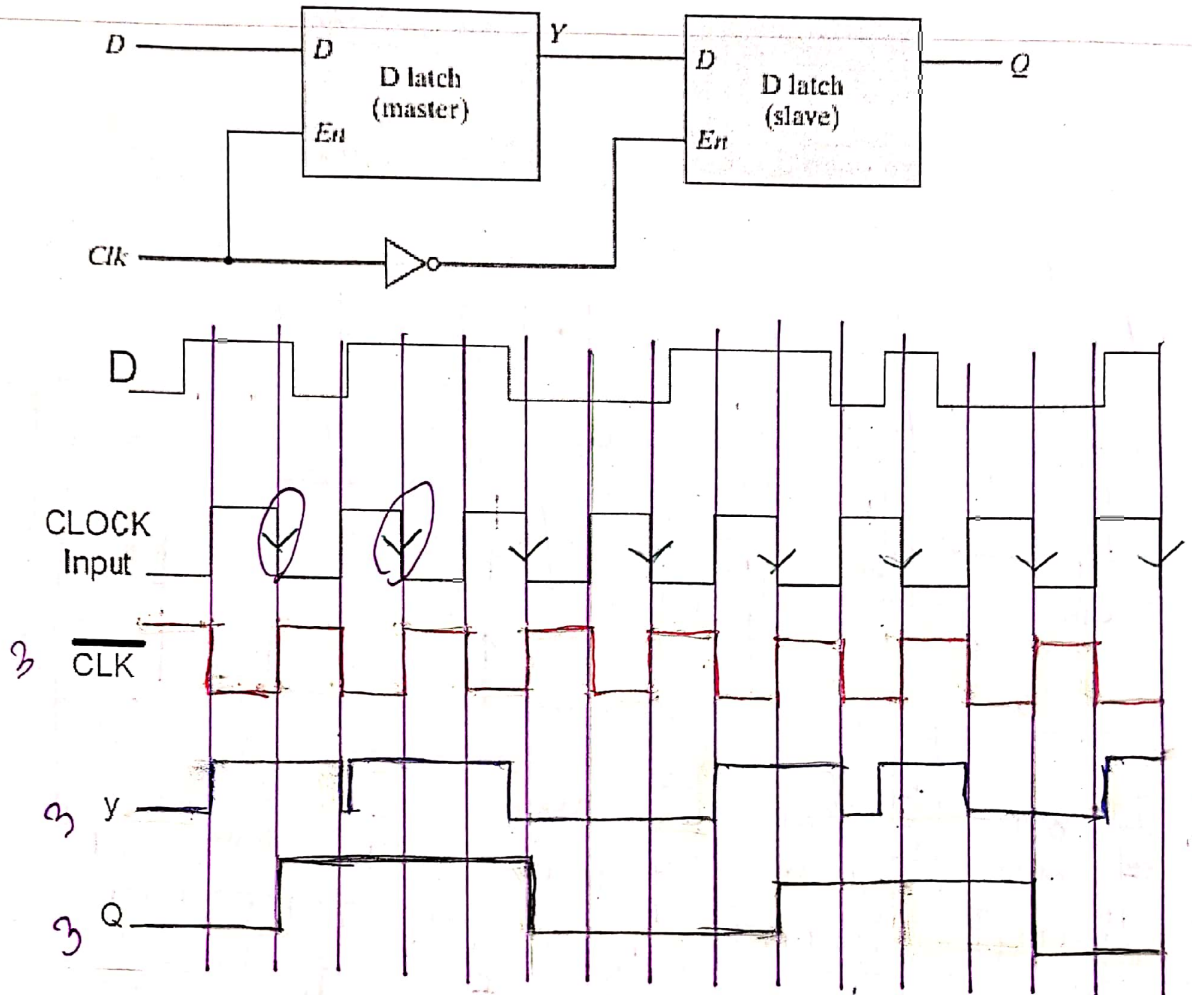
S_2	S_1	S_0	Register Operation
0	0	0	Clear to 0
0	0	1	Parallel load
0	1	0	Shift Right
0	1	1	No Change
1	0	0	Compliment Of Old State
1	0	1	Shift Left
1	1	0	Compliment of Parallel load
1	1	1	Clear to 1



Question 6 [10 Marks]

- i. Complete the timing diagram for the output $\overline{CLK}$, Y and Q by considering that you are applying given CLK and D as an input to the circuit given below where En is active high enable

Note: Highlight the part of your timing diagram that indicates that it's a Master Slave Flip flop



Its a negative ^{Edge} triggered Master Slave Flip flop.

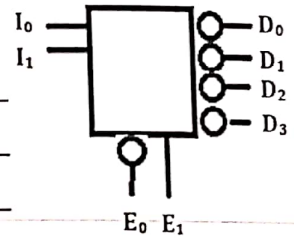
As indicated at "CLOCK INPUT", "FINAL OUTPUT" → Q is updated when clock Moves from 1 → 0 (green arrow)



Question 7 [5 + 5 Marks]

- i. Produce table & expressions for 2x4 Decoder given below?

E_1	$\bar{E}_0$	I_1	I_0	D_0	D_1	D_2	D_3
0	x	x	x	1	1	1	1
x	1	x	x	1	1	1	1
1	0	0	0	0	1	1	1
1	0	0	1	1	0	1	1
1	0	1	0	1	1	0	1
1	0	1	1	1	1	1	0



$$D_0 = \bar{E}_1 \cdot \bar{E}_0 \cdot \bar{I}_1 \cdot \bar{I}_0 = \bar{E}_1 + \bar{E}_0 + \bar{I}_1 + \bar{I}_0$$

$$D_1 = \bar{E}_1 \cdot \bar{E}_0 \cdot I_1 \cdot \bar{I}_0 = \bar{E}_1 + \bar{E}_0 + I_1 + \bar{I}_0$$

$$D_2 = \bar{E}_1 \cdot E_0 \cdot \bar{I}_1 \cdot \bar{I}_0 = \bar{E}_1 + E_0 + \bar{I}_1 + \bar{I}_0$$

$$D_3 = \bar{E}_1 \cdot E_0 \cdot I_1 \cdot \bar{I}_0 = \bar{E}_1 + E_0 + I_1 + \bar{I}_0$$

- ii. Implement following function using 4x1 multiplexer?
- $F(A, B, C) = \sum (1, 2, 5, 8, 10, 15)$

A	B	C	D	F
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

$$\Rightarrow F = C'D + CD' = C \oplus D$$

$$F = C'D$$

$$F = C'D' + CD' = (C' + C)D' = D'$$

$$F = CD$$

